Chapter 9. The Origins of Graph Theory

Dots and Lines

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In 18th-century Königsberg seven bridges connected the land masses around the river Pregel. In 1736 Euler asked whether one can walk through the town, starting and ending at the same place, crossing every bridge exactly once.

9.1 The notion of graphs. Eulerian trails

Motivation. Euler saw that only the pattern of connections matters, not the geography: land masses become dots and bridges become lines. Whether a closed walk can use every line exactly once then depends only on the parity of the number of lines at each dot.

A **graph** consists of finitely many **vertices** (dots) and finitely many **edges** (lines), each joining two vertices. The **degree** of a vertex is the number of edge-ends at it. Graphs may have multiple edges and **loops** (edges from a vertex to itself); a graph with neither is **simple**.

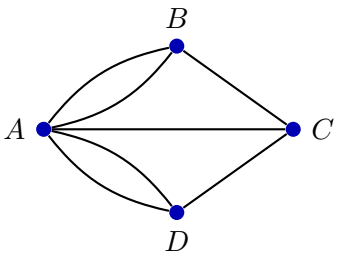


Figure. *The Königsberg bridges as a graph: four land masses, seven edges. All four degrees $(5, 3, 3, 3)$ are odd, so no closed walk uses every bridge once (Theorem 9.2).*

A sequence of distinct edges $e_1 e_2 \cdots e_k$ is a **trail** if the end of e_i is the start of e_{i+1} (it can be drawn without lifting the pencil); a **walk** may repeat edges. A trail is **closed** if it ends where it starts, **Eulerian** if it uses all edges, and a **path** if it visits no vertex twice.

Definition 9.1 (Connected)

G is **connected** if any two vertices are joined by a path.

A **subgraph** H of G has vertices and edges among those of G. It is **induced** if it contains all edges of G between its vertices. The **connected components** are the maximal connected subgraphs; u, v lie in the same component iff a path joins them.

Theorem 9.2 (Euler)

A connected graph has a closed Eulerian trail iff all its degrees are even.

Proof.

Step 1 (only if). Follow a closed Eulerian trail W from its start S. Each visit to a vertex $A \neq S$ enters and leaves along two new edges, and W uses every edge, so $\deg A = 2a$ if A is visited a times. S has the first and last edge plus $2t$ edges for t intermediate visits: $\deg S = 2t + 2$.

Step 2 (a first closed trail). Assume all degrees are even. From any vertex S walk along unused edges. Whenever we enter a vertex other than S, an unused edge is available to leave it (even degree), so we can only get stuck at S; as G is finite we do return to S, giving a closed trail C_1.

Step 3 (a vertex with unused edges). If C_1 is not all of G, some vertex V of C_1 has an edge not in C_1. Otherwise some vertex A is not on C_1, and a path from A to C_1 (connectedness) enters C_1 first at a vertex V through an edge not in C_1, a contradiction.

Step 4 (splicing). Removing the edges of C_1 keeps all degrees even. Starting at V as in Step 2, find a closed trail C_2 among the remaining edges. Walk along C_1 to V, around C_2, and finish C_1: this is a longer closed trail. Repeat; as there are finitely many edges, we end with a closed Eulerian trail. ■

Corollary 9.3

A connected graph G has an Eulerian trail from S to a different vertex T iff S and T have odd degree and all other degrees are even.

Proof. Add a new edge ST to get H. Eulerian trails of G from S to T correspond to closed Eulerian trails of H (close up with the new edge, or cut it out). The degrees of H are those of G with $\deg S$, $\deg T$ increased by 1. Apply Theorem 9.2 to H. ■

Theorem 9.4 (Handshake lemma)

In a graph without loops, the number of vertices of odd degree is even.

Proof. Each of the e edges contributes 1 to the degree of each of its two endpoints, so $d_1 + \cdots + d_n = 2e$ is even. Hence the number of odd summands is even. ■

9.2 Hamiltonian cycles

Motivation. Visiting every *vertex* once is much harder than using every edge once: no efficient test is known, and the problem is NP-complete. Still, enough edges force a Hamiltonian cycle, and Dirac's degree condition is the classic sufficient criterion.

A **cycle** is a closed trail that repeats no vertex except start = end; a cycle with k vertices has k edges. A **Hamiltonian cycle** (**path**) is a cycle (path) through all vertices. Example: seating guests around a round table so that neighbours know each other is a Hamiltonian cycle in the acquaintance graph.

Deciding whether a graph has a Hamiltonian cycle is **NP-complete**. A cycle is easily verified once exhibited, but no algorithm polynomial in n is known to certify that none exists; checking all $(n-1)!$ seatings is far too slow (Chapter 20 of the book).

Theorem 9.5 (Dirac)

Let $n \geq 3$ and let G be a simple graph on n vertices with all degrees at least $n/2$. Then G has a Hamiltonian cycle.

Proof.

Step 1 (connected). A component with fewer than $n/2$ vertices would contain vertices of degree $< n/2$, and if G were disconnected, some component would have fewer than $n/2$ vertices.

Step 2 (a saturated counterexample). Suppose G has no Hamiltonian cycle. Add edges as long as possible without creating one. The resulting G' still has degrees $\geq n/2$ and no Hamiltonian cycle, but adding any missing edge creates one. G' is not complete, since K_n ($n \geq 3$) is Hamiltonian.

Step 3 (a Hamiltonian path). Take non-adjacent x, y in G'. Adding xy creates a Hamiltonian cycle through xy, so G' has a Hamiltonian path $x = z_1, z_2, \ldots, z_n = y$.

Step 4 (pigeon-hole). Let $A = \{i \in \{2, \ldots, n-1\} \mid xz_i \in E\}$ and $B = \{i \in \{2, \ldots, n-1\} \mid z_{i-1}y \in E\}$. Since $x \not\sim y$, $|A| = \deg x \geq n/2$ and $|B| = \deg y \geq n/2$ (shift the indices of y's neighbours by one). Both lie in an $(n-2)$-element set, so $|A| + |B| \geq n > n-2$ and some $i \in A \cap B$ exists.

Step 5. Then $x\, z_2 \cdots z_{i-1}\, y\, z_{n-1} \cdots z_i\, x$ is a Hamiltonian cycle in G', a contradiction. ■

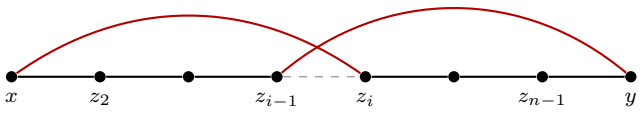


Figure. *Dirac's theorem: if $x \sim z_i$ and $z_{i-1} \sim y$ (red), drop the path edge $z_{i-1}z_i$ (dashed) to obtain the Hamiltonian cycle $x\, z_2 \cdots z_{i-1}\, y\, z_{n-1} \cdots z_i\, x$.*

9.3 Directed graphs

Motivation. One-way streets turn graphs into digraphs. Euler's criterion becomes "in-degree = out-degree". Tournaments (orientations of complete graphs) always have a Hamiltonian path, and a Hamiltonian cycle exactly when every player can be reached from every other.

In a **directed graph** (**digraph**) each edge has a direction, and trails must follow it. G is **strongly connected** if for all vertices a, b there is a directed path from a to b. The **in-degree** (**out-degree**) of a vertex is the number of edges ending (starting) there. G is **balanced** if in-degree = out-degree at every vertex. A vertex is **isolated** if both degrees are 0.

Theorem 9.6

A digraph without isolated vertices has a closed Eulerian trail iff it is balanced and strongly connected.

Proof.

Step 1 (necessity). A closed Eulerian trail leaves each vertex as often as it enters it, so G is balanced. Following the trail from any vertex reaches any other (every vertex has an edge on it), so G is strongly connected.

Step 2 (sufficiency) [Supplement]. We repeat the proof of Theorem 9.2 with directed edges. Starting at S, walk along unused edges in their direction. Balance means that each time we enter a vertex $\neq S$, an unused outgoing edge remains, so we can only stop at S, giving a closed trail C_1. If C_1 misses an edge, some vertex V on C_1 has an unused edge. Otherwise take a vertex A off C_1 (one exists, as there are no isolated vertices) and a directed path from C_1 to A (strong connectivity); its first edge leaving C_1 starts at such a V. Removing C_1 keeps the graph balanced, so a closed trail C_2 from V exists, and splicing C_2 into C_1 at V gives a longer closed trail. Repeat until all edges are used. ■

A simple graph is **complete** if all pairs of distinct vertices are adjacent ($\binom{n}{2}$ edges, K_n). Orienting every edge of K_n gives a **tournament**: the results of a round robin, with ij an edge if i beat j. There are $2^{\binom{n}{2}}$ tournaments on n labelled vertices.

Theorem 9.7

Every tournament has a Hamiltonian path.

Proof. Induction on n; $n \leq 2$ is clear. Remove a vertex V; by induction $T - V$ has a Hamiltonian path $h_1 h_2 \cdots h_{n-1}$. If $h_i V$ and $V h_{i+1}$ are edges for some i, insert V between h_i and h_{i+1}. Otherwise the edges between V and the path never switch from "into V" to "out of V". So there is k with $V h_j$ an edge for $j \leq k$ and $h_j V$ an edge for $j > k$. Then $V h_1$ is an edge (if $k \geq 1$, put V first) or $h_{n-1} V$ is an edge (if $k = 0$, put V last). ■

Theorem 9.8 (Camion)

A tournament has a Hamiltonian cycle iff it is strongly connected.

Proof.

Step 1 (only if). A Hamiltonian cycle gives a directed path between any two vertices.

Step 2 (a cycle exists). Let T be strongly connected. If T had no cycle, then $xy, yz \in E$ would force $xz \in E$ (otherwise zx closes a 3-cycle). So T would be transitive, and its vertices could be listed with every edge pointing to the right; such a tournament is not strongly connected (for $n \geq 2$).

Step 3 (a longest cycle). Let $C = y_1 \cdots y_k$ be a longest cycle and suppose it misses a vertex. By strong connectivity some edge leaves C, say $y_1 x$ with $x \notin C$. If $xy_2 \in E$, then $y_1 x y_2 \cdots y_k$ is longer, so $y_2 x \in E$; inductively $y_i x \in E$ for all i.

Step 4. Let Z be the set of $z \notin C$ with $y_1 z \in E$; by Step 3, $y_i z \in E$ for all i and all $z \in Z$. Strong connectivity gives an edge zt with $z \in Z$, $t \notin Z$. Then $t \notin C$, and $t \notin Z$ means $ty_1 \in E$. So $z\, t\, y_1 y_2 \cdots y_k$ (preceded by $y_k \to z$) is a cycle longer than C, a contradiction. ■

9.4 The notion of isomorphisms

Motivation. Two drawings may describe the same graph, and two different labellings may hide the same shape. Isomorphism makes "same up to relabelling the vertices" precise.

Two graphs on the same labelled vertex set are **identical** if every pair of vertices is joined by the same number of edges in both, however they are drawn.

Definition 9.9 (Isomorphism)

Graphs G, H are **isomorphic** if there is a bijection f from the vertices of G onto those of H such that X, Y and $f(X), f(Y)$ are joined by the same number of edges, for all X, Y. Such f is an **isomorphism**.

Example 9.10

For the two 4-vertex graphs in the book's Figure 9.8, $f(A) = H$, $f(B) = E$, $f(C) = F$, $f(D) = G$ is an isomorphism. The graphs are not identical (the first contains AB), but they are the same as unlabelled graphs.

Isomorphisms preserve degrees, numbers of cycles, numbers of paths of each length, and so on. Different degree multisets therefore prove non-isomorphism, but equal degree multisets do not prove isomorphism. No efficient general isomorphism test is known (checking all $n!$ bijections is too slow), and it is not known whether the problem is NP-complete; Babai's recent quasi-polynomial algorithm is a major advance.

9.5 Selected exercises

Motivation. The book exercises below combine the basic tools of the chapter (degree counting, Eulerian trails, and longest paths) to force or construct substructures.

Exercise 9.3 (A 4-cycle from many edges)

Let G be a simple graph on 10 vertices with 28 edges. Prove that G contains a cycle of length 4.

Solution.

Step 1 (cherries). Count the pairs $(v, \{x, y\})$ where $x \neq y$ are both neighbours of v. Vertex v contributes $\binom{d_v}{2}$ of them, so there are $\sum_v \binom{d_v}{2}$ pairs.

Step 2 (if there is no 4-cycle). Suppose G has no 4-cycle. If two different vertices $v \neq w$ had two common neighbours $x \neq y$, then $x, y \notin \{v, w\}$ (the graph has no loops) and $v\,x\,w\,y$ would be a 4-cycle. So every pair $\{x, y\}$ occurs in at most one cherry, and $\sum_v \binom{d_v}{2} \leq \binom{10}{2} = 45$.

Step 3 (the degrees force more). We have $\sum_v d_v = 2 \cdot 28 = 56$, and by the Cauchy–Schwarz inequality $\sum_v d_v^2 \geq \frac{1}{10}\big(\sum_v d_v\big)^2 = 313.6$. Hence

$$\sum_v \binom{d_v}{2} = \frac{1}{2}\Big(\sum_v d_v^2 - \sum_v d_v\Big) \geq \frac{313.6 - 56}{2} = 128.8 > 45,$$

contradicting Step 2. So G contains a 4-cycle. ∎

Exercise 9.6 (Two orange and two blue edges at each vertex)

Let G be a simple graph in which every vertex has degree 4. Prove that the edges of G can be coloured orange and blue so that every vertex is incident to two orange and two blue edges.

Solution. It suffices to colour each connected component H separately. All degrees in H equal 4, so H has edges, and by Theorem 9.2 it has a closed Eulerian trail $v_0, e_1, v_1, e_2, \ldots, e_m, v_m = v_0$.

Step 1 (the colouring). Each edge has two ends, so $2m = \sum_{v \in H} \deg v = 4|V(H)|$ and $m = 2|V(H)|$ is even. Colour e_i orange for odd i and blue for even i, and put $e_{m+1} = e_1$.

Step 2 (edges at a vertex). Let $v \in H$. Each edge at v has exactly one end at v (no loops) and occurs exactly once in the trail. So the four edges at v are grouped into the pairs $\{e_i, e_{i+1}\}$ for the indices $i \in [m]$ with $v_i = v$: entering v along e_i and leaving along e_{i+1} (for $i = m$, the trail closes up via e_1). There are two such indices.

Step 3 (conclusion). In each pair the indices i and $i+1$ have different parity; this holds also for the pair $\{e_m, e_1\}$, because m is even. So each pair has one orange and one blue edge, and v is incident to two orange and two blue edges. ∎

Exercise 9.23 (High minimum degree forces connectivity)

(a) Prove that a simple graph on ten vertices in which every vertex has degree at least five is connected.
(b) Generalize to simple graphs on n vertices.

Solution. We prove (b): a simple graph G on n vertices with all degrees at least $\frac{n-1}{2}$ is connected; for $n = 10$ this gives (a), as $5 \geq \frac{9}{2}$.

Step 1. Let $u \neq v$ be vertices that are not adjacent. Their neighbourhoods $N(u), N(v)$ are subsets of the $(n-2)$-element set $V(G) \setminus \{u, v\}$, and

$$|N(u)| + |N(v)| \geq n - 1 > n - 2.$$

So $N(u) \cap N(v) \neq \emptyset$: some w is adjacent to both, and $u\,w\,v$ is a path.

Step 2. Hence any two vertices are joined by a path of length at most 2, and G is connected.

The bound is sharp: for even n, two disjoint copies of $K_{n/2}$ have all degrees $\frac{n}{2} - 1$ and are disconnected. ■

Exercise 9.46 (A long cycle in a regular graph)

Let $k \geq 2$ and let G be a simple graph in which every vertex has degree k. Prove that G contains a cycle of length at least $k + 1$.

Solution.

Step 1 (a longest path). G has finitely many paths and at least one edge, so let $P = v_0 v_1 \cdots v_m$ be a path with the largest number of vertices.

Step 2 (neighbours of an end). Every neighbour of v_0 lies on P: if $u \notin P$ were adjacent to v_0, then $u\, v_0\, v_1 \cdots v_m$ would be a longer path. The k neighbours of v_0 are distinct vertices among $v_1, \ldots, v_m$ (G is simple), so the largest index j with v_j adjacent to v_0 satisfies $j \geq k \geq 2$.

Step 3 (the cycle). Then $v_0 v_1 \cdots v_j v_0$ is a cycle (its vertices are distinct and $j \geq 2$), of length $j + 1 \geq k + 1$.

The proof only uses that all degrees are at least k. The bound is attained by K_{k+1}. ■